

CHENAIT OUSSAMA

Senior Software Engineer | Game Engine & Systems Developer | Full-Stack & Mobile Applications

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PROFILE SUMMARY

Highly accomplished Senior Software Engineer and Game Developer with over 6 years of experience across low-level systems emulation, custom game engine development, and advanced Unreal Engine architecture. Skilled full-stack web developer and mobile application developer, delivering production-ready web applications, mobile apps, and enterprise business systems for companies and private clients. Proficient in bridging complex C++ simulations with scalable full-stack web technologies, mobile platforms, responsive interfaces, and optimized database frameworks. A versatile problem solver dedicated to writing clean, maintainable code and building immersive, reliable software across desktop, web, and mobile.

WORK EXPERIENCE

Lead Systems & Game Engineer

2021 - Present

Freelance / Self-Employed

- Architected and engineered a custom low-level PlayStation 5 emulator from scratch, simulating hardware subsystems to run 2D and demanding 3D titles with stable resource allocation.
- Designed and developed a highly interactive open-world sandbox game within Unreal Engine, programming procedural animal simulation systems and complex animation states.
- Built a proprietary 3D modeling and rendering application (Blender-like architecture) and a custom C++ game engine, mastering graphics pipeline processing and runtime execution.
- Developed and deployed 17+ interactive, production-ready full-stack web applications using Next.js, React, PHP, and optimized relational SQL databases.
- Designed and built mobile applications for clients and internal projects, delivering cross-platform apps integrated with backend APIs and databases.
- Developed custom business/enterprise systems for companies, including internal management tools, automated workflows, and database-driven administration platforms.

FiveM Infrastructure & Multiplayer Developer

2020 - 2026

Private Multiplayer Servers

- Engineered and optimized multiplayer environments and dedicated server frameworks for the FiveM platform using Lua and JavaScript.
- Developed custom high-performance server-side infrastructure and advanced synchronization algorithms to handle massive real-time player data with minimal latency.
- Programmed bespoke gameplay scripts, integrated backend web systems with game databases via optimized SQL queries, and resolved severe runtime memory leakage issues to sustain peak server performance.

TECHNICAL SKILLS

- Programming & Engine Architecture: C++, Custom Game Engine Development, Low-Level System Emulation, Unreal Engine, Unity, Godot
- Web Development: Full-Stack Web Development, React.js, Next.js, PHP, Responsive UI, Web Application Development
- Mobile Development: Mobile Application Development, Cross-Platform Apps, API Integration
- Enterprise & Business Systems: Company/Enterprise Systems Development, Internal Management Tools, Workflow Automation, Database-Driven Platforms
- Databases: Advanced SQL, Relational Database Design, Query Optimization
- Multiplayer & Server Engineering: FiveM Server Engineering, Lua, JavaScript, Real-Time Synchronization, Server-Side Infrastructure
- Performance & Optimization: Memory Leak Resolution, Runtime Debugging, Resource Management, Graphics Pipeline Optimization

CORE SKILLS

C++, Unreal Engine, Unity, Godot, Game Engine Development, Low-Level Programming, System Emulation, Real-Time Systems, Full-Stack Development, React.js, Next.js, JavaScript, PHP, Web Application Development, Mobile Application Development, SQL, Database Design, API Integration, Enterprise Systems Development, Multiplayer Systems, Server-Side Development, FiveM, Lua, Performance Optimization

EDUCATION

Bachelor in Computer Science

University of Science and Technology Houari Boumediene (USTHB) | 2023 - 2026

Technical Mathematics – Academic Branch Track

UDEMY | 2023 - 2026